

Statistics Hybrid Practice Set – S10 (L6)

Topics Covered

- Population
- Sample
- Mean
- Median
- Mode
- Range
- Population Variance
- Population Standard Deviation
- Quartiles (Q1, Q2, Q3)
- IQR
- Outlier Detection
- **Frequency Distribution (New) ★**

Instructions

- Solve on **paper first**.
 - Then solve the **same questions in Excel**.
 - Show your calculations.
 - No calculator.
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Q1. Identify Population and Sample

A supermarket has **22,000 customers**. It surveys **750 customers**.

Paper Task

- Population = ?
- Sample = ?

Excel Task

Total Customers	Surveyed Customers
22000	750

Q2. Calculate the Mean

Value
24
28
32
36
40

Paper Task

- Find the Mean.

Excel Task

- Verify your answer in Excel.
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Q3. Find the Median

Value
39
27
51
33
45

Paper Task

- Find the Median.

Excel Task

- Verify your answer.
-

Q4. Find the Mode

Value
15
17
17
19
21
23

Paper Task

- Find the Mode.

Excel Task

- Verify your answer.
-

Q5. Complete Statistical Analysis

Value
18
24
30
36
42

Paper Task

Calculate:

- Mean
- Range
- Population Variance
- Population Standard Deviation

Excel Task

Verify all four answers.

★ Q6. Frequency Distribution (New)

The following marks were obtained by **20 students**.

Marks
41
45
47
49
52
55
57
58
61
63
66
68
71
73
75
77
82
84
88
91

Create a **Frequency Table** using the following class intervals:

Class Interval	Frequency
40–49	?
50–59	?
60–69	?
70–79	?
80–89	?
90–99	?

Excel Task

Create the same frequency table.

★ Q7. Frequency Interpretation

Using your frequency table from **Q6**, answer:

1. Which class interval has the **highest frequency**?
 2. Which class interval has the **lowest frequency**?
 3. How many students scored **70 or above**?
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Q8. Outlier Detection

Dataset

Value
25
27
29
31
33
35
37
39
90

Paper Task

Find:

- Q1
- Q3
- IQR
- Lower Fence
- Upper Fence

Is **90** an outlier?

Excel Task

Verify your answer.

Q9. Real Data Analyst Scenario

A company wants to know **how many employees fall into different salary ranges** instead of looking at individual salaries.

- Which statistical tool is the **best** choice?

- A. Mean
- B. Frequency Table
- C. Mode
- D. Standard Deviation

Explain your answer in one sentence.

Q10. Mixed Business Case (No Formula Question)

- A manager collected the following daily sales:

120, 122, 121, 119, 350

Answer the following:

1. Which measure of central tendency is **most reliable** for this dataset?
2. Is there likely an **outlier**?
3. Which method would you use to confirm it?
4. Would you report the **Mean** or **Median** to the manager? Why?